AIRSTREET · RISK REGISTER

AirStreet risk register and hazard analysis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	AST-RSK-001	Version	0.1
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/airstreet		
Source	docs/09-risk.md @ aa55b15		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA (kit 1.9.1 upgrade)

AirStreet risk register and hazard analysis

The largest project risks are that no partner with a street-level NO2 reference station is found, and that the field NO2 accuracy after calibration is worse than the 5 ppb target; the largest hazards are work at height beside traffic and a node or part falling from the pole. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be found) to 5 (unlikely to be found).

> **Safety:** Mounting on a street pole is work at height next to traffic, beside a pole that carries mains voltage inside. Install only with the pole owner's written permission, by trained crews from a work platform or a secured ladder with a second person and traffic management, and never open the pole's access door. The FieldNode core holds a lithium iron phosphate cell; the NO2 cell holds an acid electrolyte.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No collocation partner	No city or agency with a street-level NO2 reference station agrees to host nodes and share data	NO2 cannot be calibrated; the method is unproven	3	5	Project lead	Approach the first candidate (TCEQ Dallas-Fort Worth network with a university or city partner, AST-DEC-001); keep a list of alternates
P2	Field NO2 accuracy worse than 5 ppb hourly	Calibration residual, drift and street transfer bias larger than assumed (AST-CAL-001 C)	PR-002 not met; maps can rank streets only coarsely	3	4	Design lead	Collocation statistics at TRL 4; the diffusion tube cross-check (open decision 2)
P3	PM2.5 wrong in humid weather	Water uptake above 85 % RH (PR-001)	Overestimated PM on humid nights and in humid climates	4	3	Design lead	Humidity flags (F2); collocation in the humid season; a switched inlet heater as a later idea (AST-OPT-001)
P4	Pole owner refuses the load or the mount	3.77 kg and 125 N of wind load at 3.3 m; insurance	No sites	2	4	Project lead	Give the owner the load and wind figures; no drilling or pole power (PR-011, PR-018)
P5	FieldNode does not accept the adapter plate mount	AirStreet leaves off FieldNode's back plate (AST-DDR-002)	A redesign of the mount	2	3	Project lead	Cross-repo request already approved (AST-DEC-001)
P6	Cost over the value-engineering target	NO2 cell and front end at about USD 150 together	USD 17.50 over the USD 285 target (PR-015)	5	2	Design lead	Open two-channel front end (TRL 4 work on hold); quotes from distributors
P7	NO2 cell price or supply	One maker in practice; price not quoted	Cost and lead time	3	3	Design lead	Quote at TRL 4; name an equivalent class of cell
P8	Theft or vandalism	Device in public reach at 3 m	Lost nodes and data	2	3	Partner organization	Inlet height 3.0 m; no exposed fixings below the pod; owner's inspection rounds
P9	The 2030 EU limit makes	Directive (EU) 2024/2881 lowers	Classification is against the wrong value	4	2	Project lead	Open decision 4

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
	PR-004 out of date	the NO2 limit to 20 ug/m3					

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Fall from height	Installing or exchanging the pod at 3 m from a ladder	Serious injury	5	2	Work platform or secured ladder with a second person; the pod exchange is done whole, with no sensor work at height (PR-016)	Low with trained crews
H2	Struck by traffic	Work beside the kerb	Serious injury	5	2	Traffic management as local rules require (AST-REG-001 S2); work from the footway side	Low
H3	Falling node or part onto the footway	Band slip, a loose screw, a cracked printed part	Injury to the public	4	2	Two bands with a slip factor of 25 (AST-CAL-001 [G2]); nyloc nuts and inserts; re-torque at the first pod exchange (S25); safety stops in the build plan	Low; insert pull-out to check at TRL 4
H4	Electric shock from the pole	Opening the pole's access door or tapping pole power	Death	5	1	Never open the pole; AirStreet runs only on FieldNode's solar and battery supply (PR-018)	Very low
H5	Lithium cell fire or venting	Damage, overcharge, charging below 0 degC or above the maker's limit, a hot site without the sun shield	Fire at height	4	1	Fused, protected LiFePO4 cell; charge temperature limits in FieldNode; sun shield above 45 degC (AST-DDR-002)	Low
H6	Acid electrolyte	Crushing or heating an electrochemical cell	Burns to skin or eyes	3	1	Cells fitted and replaced on the ground, returned for disposal (AST-REG-001 S3)	Very low
H7	Cuts	Band ends and cut aluminium	Minor cuts	2	3	Deburr, gloves (AST-BLD-001)	Low
H8	Misleading data	Readings outside the sensors' rated range or before calibration taken at face value	Wrong decisions about a street	3	3	Rated-range flags in every record (F2); calibration version on every published value; data stated as indicative (PR-007)	Medium until collocation

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Bands and V-saddles	Hold the node on the pole	Preload lost, node slips or turns	Node drops or faces the wrong way	Creep of printed ASA, band housing slip	4	2	3	24	Band torque figure from the maker (AST-DEC-001 item 1); re-torque at the first pod exchange; creep check at TRL 4
F2	Pod hanging screws and inserts	Hold the pod under the cross arm	Insert pulls out	Pod falls	Insert set badly; ASA softened in sun	4	2	3	24	Insert pull-out test at TRL 4 (item 8); safety stop before the first lift
F3	NO2 cell	Measure NO2	Sensitivity loss or ozone breakthrough	NO2 biased	Ozone filter used up; electrolyte dried at high temperature	3	3	2	18	Replacement every 12 months, or 18 to 24 with the site's ozone record (F3); 6-monthly collocation; flags above 35 degC and outside 15 to 85 % RH (F2)
F4	Particle sensor	Measure PM	Fan or optics fouled	PM biased low	Dust and traffic spray	3	3	3	27	Duty cycle of 60 s in 300 s; checked at each CalRig visit
F5	Radiation shield	Give air temperature and RH	Warm bias in low wind	RH low, PM correction wrong	Too little air through the plates	3	3	3	27	Side-by-side with an aspirated reference at TRL 4 (PR-005)
F6	Pod	Keep rain and insects out of the sensors	Water or insects reach the sensors	Sensor failure or false PM	Driven rain under the drip lid; nests	3	3	2	18	Mesh aperture to confirm (item 9); inspection at the pod exchange
F7	Sensor cables and glands	Carry power and data to FieldNode	Water in a gland or plug	Lost data, corrosion	Gland not tightened; plug not seated	3	2	2	12	Gland check in the pod exchange steps; uplink check before leaving the pole
F8	Front end and ADC	Hold the bias and read WE and AE	Bias lost on a power dip	Hours of settling after each dip	Port B not always on	3	2	3	18	FieldNode's confirmation of the always-on port B (AST-DEC-001 action 4)
F9	FieldNode core	Power and radio	Cell empty in a long cold spell	Gap in data	No charging below 0 degC (S4)	2	2	1	4	Partner climate data; store and forward 7 days

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F10	Adapter plates	Carry the FieldNode enclosure and panel bracket	Crack at a lightening window corner	Enclosure loose	Vibration; window corner radius too small	3	1	3	9	5 mm corner radius and 5 mm of metal round every hole (AST-DDR-003)